

Forces and Motion: Roller Coaster Fun!

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Lesson # 22

Forces and Motion: Roller Coaster Fun!

Teacher Section

Purpose

The purpose of this activity is to provide students an opportunity to learn and apply science process skills and conceptual understanding of forces and motions, and work in cooperative groups to problem solve.

Objectives

The students will be able to:

- Create a technical drawing.
- Investigate the variables and speed (e.g. ramp height/length/surface and mass of object).
- Describe how forces affect the motion of objects.
- Describe the relative positions and movements of objects using points of reference.
- Design and build a virtual model and a 3-D model of a roller coaster.
- Communicate ideas using drawings, models, and written expression.
- Investigate, explain, and demonstrate knowledge of Newton's Laws of Motion, energy (e.g. kinetic, potential), forces (friction, centrifugal, centripetal, gravity), and how real-world roller coasters work.
- Explain transfer of energy.
- Use the Internet to research and construct a roller coaster simulation.

Grade Level

This activity is designed for Grades 5th -7th.

National Science Education Standards

K-12 Unifying Concepts & Processes:
Evidence, models and explanation

Grades 5-8:
Science as Inquiry:
Abilities necessary to do scientific inquiry
Understandings about scientific inquiry
Physical Science:
Motions and forces
Transfer of Energy
Science and Technology:
Abilities of technical design
Understanding about science and technology
Science in Personal & Social Perspectives:
Science & technology in society

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Including All Students

This activity addresses multiple modalities: Kinesthetic (3-D models, drawing and writing), Visual and Auditory (computer simulation and research, technical drawing, group interactions).

Safety

- During the model-building activity, students shouldn't stand on furniture unless they ask permission.
- Students should handle scissors, and rulers properly.
- While using the Internet, students should not "surf" off of the assigned web pages.

Questions To Ask Along The Way

- What causes your coaster to slow down or resist movement? What can you do to over come the resistance?
- How does the height (steep or not so steep) of the hills affect the coaster? Is the placement of hills important?
- Can all of hills be the same height? Why or why not?
- What kinds of curves (sharp or gradual) work best?
- What are the effects of loops? Is the placement (end, middle, end) of the loop important?

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Materials

For each group:

- Computer with access to the Internet
- Student handouts
- Polyester pipe insulator tube (6ft., cut lengthwise)
- Poster paper
- Colored pencils
- Notebook paper
- Marbles
- ¼ inch graph paper
- Meter sticks or tape measure
- Pencils

Preparation

- Collect pictures and postcards to include in the Engage part of the lesson. A short video clip of a roller coaster ride is also effective. You can also ask the students to bring related amusement park pictures, post cards, etc.
- Purchase pipe insulators at any plumbing supply store (about $1.50 per group). The 6 ft. ¾" diameter insulators should be cut the length of the tube exposing the groove prior to the activity. To make longer coasters, use tape to attach another length of tubing.
- The teacher should print out the student assignment sheets.

Procedure

Engage

- The teacher introduces the topic with the question: *How many of you enjoy amusement park rides?* Monitor responses to make sure all students are engaged. Ask students to describe their favorite rides.
- After the class discussion, students should create a **KWWL** chart (what I **k**now, what I **w**ant to know, **w**here I might find that information, and what I **l**earned) in their science notebooks and complete the K, W and W sections. They may work in groups to brainstorm and share prior knowledge and experiences. After students complete the chart, the class shares their ideas.
- The teacher asks leading questions. For example: *What questions do you have about how roller coaster works? What makes the ride so thrilling, fun, exciting?* Allow students to make predictions and hypothesize based on prior experiences and knowledge. The teacher should create a class chart on large poster paper (class KWWL document). Direct students to respond to the **K**, **W**, and **W** headings on class chart.

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Procedure

Explore:

- The students will work cooperatively, in groups of three, to complete the Internet research component of this activity. Instruct students to acknowledge vocabulary during the research.
- After the Internet Info Search, direct student groups to use graph paper to make a rough "technical drawing" of a marble roller coaster model using pencils and the outlined design specifications (specific length of the coaster, one loop, one hill, one curve, marble has to stay on the track/safety issues). This will be their hypothesis based on background information acquired from the Internet research, prior knowledge and experiences.
- Groups should develop and record a hypothesis (if my first hill is so high, then…., for example). This process may require more time with some groups. The students should use their draft to build a 3D model of their coaster and test it at least three times.
- After testing, students are allowed to revise their drawing and models, if necessary. Then, the students can test their coasters again and report on the result (better/worse than first design, variables changed, future modifications, etc.). Consequently, the students may continue to use the Internet to create, test, and revise their roller coaster. The teacher should look for and lead students to deal with misconceptions.

Explain:

- The teacher will ask the students: How do forces and motion influence roller coaster rides? Follow this with a class discussion. Revisit the L of the KWWL chart, "What did you learn?" Lead a discussion and listen for use of vocabulary, concepts and misconceptions. Then, provide activities and opportunities to eliminate any misconceptions.

Extend:

- When finished, each group will present their virtual creations, blue prints, and 3-D model to the class. The students will demonstrate conceptual understanding of science terms (ex. forms of energy, Newton's laws, forces, and motion) by correctly using the terms in their presentation. Constructed three-dimensional models must meet specifications. The marble/rider must stay on the track from beginning to end of the ride/tubing.

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Procedure

Evaluate:

- Note: Some groups may need and be given extra time to revise their models before or after the presentation.
- **Discussion/Presentation Questions:** Use the following questions to prompt students and assess informally or formally and deal with misconceptions. These questions may be used to assess individuals or groups (written or oral questions).
  1. What forces cause the motion of the roller coaster?
  2. At what point does the gravitational potential energy and kinetic energy transform from one to the other.
  3. Identify the point/s at which the gravitational potential energy and kinetic energy is/are the greatest on your roller coaster.
  4. Identify points where the forces are unbalanced and balanced on your roller coaster.
  5. Where might you have a sense of weightlessness and heaviness? Explain this illusion use Newton's Law s of inertia.
  6. Where on the roller coaster does the velocity increase? Why?
  7. Explain the effects of friction on the roller coaster.

Where To Go From Here

- The class will vote to determine which ride is the most thrilling roller coaster ride.
- Direct the students to write a persuasive business letter to an amusement park explaining why they should buy their roller coaster model (virtual/ three-dimensional). Incorporate six traits and expository writing model.
- The students can research the history of the roller coaster to write a news article.
- The students may create a brochure about their roller coaster design or Power Point presentation incorporating graphics and pictures.

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Assessment Suggestions

- The teacher will use a rubric to assess models, blueprints, and letters. The teacher will assess science notebook responses looking for misconceptions, the correct use of vocabulary and concepts. Students must use the correct scientific terminology in all of their writing and oral presentations. They will be assessed individually.

Each student can be assessed on the following items:

Virtual Roller Coaster: (demonstrate participation)
Should demonstrate completion of exploration and meet criteria.
Should be printed and downloaded on a disc for presentation.

Technical Drawing: Should be neat, legible, labeled, in color on graph paper.

Three-Dimensional Models: (demonstrate participation)
Should be functional according to specifications.
Should be a 3-dimensional model of the technical drawing.
Marble should not leave the track during demonstration.

Persuasive Letter: Each student should a letter. The letter should be graded using the six traits of writing. The letter should demonstrate criteria of expository persuasive writing. Ideas should be written in paragraphs in a business letter format.

Student Notebooks: Assess notebooks and KWWL charts. Check for use of vocabulary, concepts, and misconceptions. Look for the use of diagrams, drawing, and labels. All entries should be dated and organized.

Reflection: Students will complete the self-assessment form. This may be done in the science notebook. Discuss the value of this piece.

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KWWL

Student Section

How does a roller coaster work?			
What do you **know**?	What do you **want** to know?	**Where** do you find answers?	What did you **learn**?

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Lab Guide Sheet

Student Section



Dear Student Team Members,

Your team has been assigned the awesome task of creating a" thrilling" roller coaster for a famous amusement park. To do the best job, you must complete an Internet Info Search first. A list of required and suggested web sites is attached.

Your final product must meet the following specifications:

- The roller coaster should demonstrate your knowledge and understanding of forces, motion, energy, and Newton's Laws.
- The final roller coaster must possess at least one loop, one curve, and a hill.

Your team has 90 minutes to explore, design, and construct your coaster. Your team must, in addition, create a technical drawing, and a 3-D model of your coaster. Finally, your team will present your product and ideas to the class. Afterwards, your team will create a name for your coaster, and write a letter to Disney persuading them to buy your coaster.

Sincerely,
Your teacher

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Team Members:

1.
2.
3.

Questions to Ponder:

Think about the following questions as you design and construct your model.

1. What causes your coaster to slow down or resist movement? What can you do to over come the resistance?
2. How does the height (steep or not so steep) of the hills affect the coaster? Is the placement of hills important?
3. Can all of hills be the same height? Why or why not?
4. What kinds of curves (sharp or gradual) work best?
5. What are the effects of loops? Is the placement (end, middle, end) of the loop important?

Safety while building your model:

Don't stand on furniture, ask permission if necessary.
Don't place materials in your mouth.
Handle scissors, rulers, properly.

Computer Safety:

No surfing.

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Rubric

Student Section



Cooperative Group Assessment

*Name:*_____

Assess your work and your group's work in on this project.

	How did the **group** demonstrate this trait?	How did **you** demonstrate this trait?
Participation		
Creativity		
On Task		
Logical Method		
Organization		
Cooperation		

How can you improve your participation in the future?

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Internet Info Search

Student Section

The following web sites will help you answer questions about forces, motion and roller coasters.

Review each web site for information and consider who wrote the site, what their purpose is in writing it, and the credibility (accuracy) of the information.

- http://media.travel.discovery.com/ideas/themeparks/rollercoasters/buildacoaster.html
- http://www.funderstanding.com/k12/coaster/index.html
- http://www.middleweb.com/coaster.html
- http://library.thinkquest.org/2745/data/ke.htm
- http://www.isn.net/~jypsy/rcgravity.htm
- http://www.youth.net/cec/cecsci/cecsci.187.txt
- http://www.learner.org/exhibits/parkphysics/coaster.html
- http://library.thinkquest.org/26455/amuse/roller.shtml
- http://www.kidzworld.com/site/p4633.htm
- http://www.bbc.co.uk/schools/scienceclips/ages/10_11/forces_action.shtml

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Internet Info Search

Student Section

This info search will help you find information about forces and roller coasters. Your teacher has a list of web sites that will help you answer the questions.

Review each web site for information and consider who wrote the site, what their purpose is in writing it, and the credibility (accuracy) of the information.

How does the height (steep or not so steep) of the hills affect the coaster? Is the placement of hills important? How does gravity affect height?		
Web Site URL	Site 1: *Write the URL of the web site you used to answer the questions.*	Site 2: *Write the URL of the web site you used to answer the questions.*
Who created this web site?		
Why did they create it? (check all that apply)	☐ To provide factual information ☐ To influence the reader's opinion ☐ To sell a product or service ☐ I'm not sure	☐ To provide factual inform ation ☐ To influence the reader's opinion ☐ To sell a product or service ☐ I'm not sure
How credible (accurate) do you think the info is?	☐ Very accurate ☐ Somewhat accurate ☐ Not very accurate ☐ I'm not sure	☐ Very accurate ☐ Somewhat accurate ☐ Not very accurate ☐ I'm not sure
What did you learn?		

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